Chapter 1. Statistical Learning, Data and Exploration

Lectures 1–3 (overview of statistics/ML/AI, statistical learning, data as probabilistic evidence, assumptions, data structures, EDA)

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1 Statistics, Machine Learning, AI and Data Mining

1.1 Where are we? A statistician's view

- **Statistics**: modeling and analyzing the data-generating process under uncertainty.
- **Statistical learning**: estimating $P(Y \mid X)$ from limited samples.
- **Machine learning**: optimization for prediction.
- **Data mining**: discovery of patterns and structure in complex data.

AI is the broadest field (intelligent systems and decision making); ML is the subset that learns from data; deep learning is one kind of ML; statistics underpins many ML methods and models uncertainty; data science uses statistics and ML to extract insight from data.

Example: one problem, four perspectives — customer churn

AI: build an intelligent system that decides retention strategies. **ML**: learn $\hat{f}(X)$ to predict the churn probability. **Statistics**: which variables matter, and how certain are we? **Data mining**: discover hidden segments and behavioral structure. Same data, same problem, different questions.

Data are not just numbers but *traces of a hidden generative process.* Intelligence is not accuracy alone and learning is not memorization: it is an attempt to understand that process.

2 What Is Statistical Learning?

Statistical learning studies how to estimate unknown probabilistic relationships from finite data.

- **Supervised learning**: estimate $P(Y \mid X)$ from labeled data.
- **Unsupervised learning**: uncover structure in $P(X)$ without labels.

The course focuses less on algorithms as recipes and more on the principles behind them.

Definition: notation

Outcome (response, target) Y; predictor vector $X = (X_1, \ldots, X_p)$; observed data $\{(X_i, Y_i)\}_{i=1}^n$. The problem is **regression** if Y is quantitative and **classification** if Y is categorical.

Supervised learning: model $P(Y \mid X)$	Unsupervised learning: model $P(X)$
prediction: perform well on unseen data **inference**: understand how X influences Y **uncertainty**: reliability of predictions and estimates	**structure estimation**: properties of $P(X)$ **clustering**: latent groups, multimodality **representation learning**: low-dimensional summaries of variability

Learning is not just fitting — it is *generalization under uncertainty.*

2.1 Applications

The same framework (estimate an unknown relationship from data) covers medical risk prediction (probability of a heart attack), text classification (spam detection), image recognition (handwritten digits) and economic modeling (income versus demographic variables).

Example: spam data

4601 e-mails sent to one person ("George" at HP Labs, before 2000), each labeled spam or e-mail. Features: relative frequencies of the 57 most common words and punctuation marks. Words with the largest spam/e-mail difference: `george` (0.00 vs 1.27%), `you` (2.26 vs 1.27), `hp` (0.02 vs 0.90), `free` (0.52 vs 0.07), `!` (0.51 vs 0.11), `edu` (0.01 vs 0.29), `remove` (0.28 vs 0.01). Goal: a customised spam filter.

3 Data as Probabilistic Evidence

3.1 Sample versus population

Data are observations, not the population; observations are realizations of random variables, which arise from an unknown data-generating process. Data are **stochastic evidence, not absolute truth**.

Definition: the inferential setting

Observed sample $\mathcal{D}_n = \{Z_1, \ldots, Z_n\}$, population distribution $Z \sim P_0$, target quantity $\theta(P_0)$ (a center, a dispersion, a regression coefficient, ...). The sample is finite; the population is conceptual.

Sampling variability. Estimators are random because samples are random: tossing the same coin n times and repeating the experiment gives a different $\hat{p} = \bar{Z}$ each time, with $\mathrm{Var}(\hat{p}) = p(1-p)/n$.

3.2 Assumptions and identifiability

A finite data set is consistent with many possible underlying mechanisms; data alone do not uniquely determine the target, $\mathcal{D}_n \not\Rightarrow \theta(P_0)$. To make learning possible we restrict the set of possible worlds, $P_0 \in \mathcal{P}_{\text{model}}$; within this restricted class identification and valid procedures become possible.

Example: five rainy days

The last five days were Rain, Rain, Rain, Rain, Rain. Does it always rain? Did the rainy season just start? Were we unlucky this week? The same data admit several explanations; only an assumption (e.g. i.i.d. days, or a seasonal model) picks one.

Definition: data-generating process

$$Y = f(X) + \varepsilon,$$

where f is the underlying mechanism and ε captures randomness and unobserved factors.

Statistical modeling specifies a **class** of functions for f; different classes encode different structural assumptions and determine model flexibility, and they require different mathematical tools:

Function class	Example	Tools
linear models	$f(x) = x^\top \beta$	linear algebra, projections, least squares
parametric models	$f(x) = f(x;\theta)$, $\theta \in \Theta$	likelihood theory, asymptotics, Fisher information
RKHS (kernel) models	$f \in \mathcal{H}_K$	Hilbert spaces, kernels, representer theorem
nonparametric classes	Sobolev, Hölder balls	functional analysis, approximation theory

4 Data Structures and Their Implications

4.1 The i.i.d. assumption

Classical setting: $Z_1, \dots, Z_n \overset{\text{iid}}{\sim} P_0$ (randomized trials, independently sampled survey responses, repeated coin tosses). It implies the LLN, the CLT and $\sqrt{n}$-convergence, and enables classical inference: confidence intervals, hypothesis tests, t-tests and regression inference.

4.2 Beyond i.i.d.

- **Dependence**: temperature evolves gradually; if today is hot, tomorrow probably is too. Independence would require $\mathrm{Cov}(Z_t, Z_{t+1}) = 0$. With positive autocorrelation the effective sample size is smaller than n: for a stationary AR(1) with correlation ρ, $\mathrm{Var}(\bar{Z}) \approx \frac{\sigma^2}{n}\frac{1+\rho}{1-\rho}$.
- **High dimension**: classically p fixed and $n \to \infty$; in modern data $p \gg n$. When $p > n$ the linear system $Y = X\beta$ is underdetermined and OLS is not unique, so additional structure is required, e.g. sparsity $\|\beta\|_0 = s \ll n$.
- **Curse of dimensionality**: the number of points needed to cover the space grows multiplicatively in the dimension; covering $[0,1]^p$ by cubes of side ϵ needs $N(\epsilon) \asymp (1/\epsilon)^p$ cubes. High dimension implies exponential complexity, and local methods (Ch. 2) need exponentially many observations.
- **Partial observation**: we do not always observe the full outcome. With survival time T and censoring time C we record $\tilde{T} = \min(T, C)$; the study may end before the event occurs.
- **Measurement error**: the observed predictor may differ from the true one, $\tilde{X} = X + \delta$, and the outcome contains noise, $Y = f(X) + \varepsilon$; observed relationships may be distorted (attenuation, Ch. 5).

Remark: understanding data means ...

knowing what is observed, what is assumed, what is unobserved, and the limits of generalization.

Derivation: why the curse of dimensionality bites local methods

Let X be uniform on $[0,1]^p$. A cube neighborhood that captures a fraction r of the data must have side length $e_p(r) = r^{1/p}$. For $r = 0.01$ and $p = 10$, $e_p = 0.01^{0.1} \approx 0.63$: a "local" neighborhood covers 63% of the range of every coordinate, so it is no longer local. Equivalently, for n uniform points the median distance from the origin to the nearest point approaches $1/2$ as p grows.

5 Exploratory Data Analysis

5.1 Visualization

EDA visualizes the distribution of each variable, relationships between variables, structure (correlation, clusters), high-dimensional structure (projections), temporal and spatial patterns, anomalies and missing data, and model-based diagnostics.

5.2 The geometry of the data cloud

Gaussian distributions produce elliptical density contours determined by the covariance matrix. Introductory statistics studies the geometry of the cloud:

Aspect	Summary	Classical test
center (location)	mean	z-test, t-test
dispersion (spread)	standard deviation	χ^2-test, F-test
orientation (direction)	correlation	t-test for correlation

Beyond center and spread, higher moments describe the **shape**: skewness $\mathbb{E}[(X-\mu)^3]/\sigma^3$ (asymmetry) and kurtosis $\mathbb{E}[(X-\mu)^4]/\sigma^4$ (tail heaviness; 3 for the normal).

5.3 Other summaries

Quantiles (median, quartiles, percentiles), the five-number summary $(\min, Q_1, \text{median}, Q_3, \max)$, robust statistics (median, IQR $= Q_3 - Q_1$, MAD $= \text{median}\,|X_i - \text{median}\,X|$), and visual summaries (boxplots, density plots). Robust summaries have high breakdown points: one wild value can move the mean arbitrarily far but cannot move the median beyond the range of the remaining data.

Summary

- Statistical learning estimates $P(Y \mid X)$ (supervised) or the structure of $P(X)$ (unsupervised) from finite, noisy samples.
- Data are random evidence about a population; without assumptions the target is not identified.
- The model $Y = f(X) + \varepsilon$ with a chosen function class fixes the flexibility and the mathematical tools.
- Dependence, high dimension, censoring and measurement error change what can be learned; EDA examines center, spread, orientation and shape before modeling.

6 Selected exercises

Motivation. Four short derivations behind claims made in this chapter: the effective sample size under autocorrelation, the emptiness of high-dimensional neighborhoods, the robustness of the median, and the non-uniqueness of least squares when $p > n$.

Exercise 1.1 (Variance of the mean under AR(1) dependence)

Let (Z_t) be stationary with mean μ, variance $\sigma^2 > 0$ and $\mathrm{Corr}(Z_s, Z_t) = \rho^{|s-t|}$, $|\rho| < 1$ (the stationary AR(1) correlation).

(a) Show that $\mathrm{Var}(\bar{Z}_n) = \dfrac{\sigma^2}{n}\Big[1 + 2\sum_{h=1}^{n-1}\Big(1 - \dfrac{h}{n}\Big)\rho^h\Big]$.

(b) Show that $n\mathrm{Var}(\bar{Z}_n) \to \sigma^2 \dfrac{1+\rho}{1-\rho}$.

(c) Interpret $n_{\text{eff}} = n(1-\rho)/(1+\rho)$.

Solution [Supplement].

(a) By bilinearity, $\mathrm{Var}\big(\sum_{t=1}^n Z_t\big) = \sum_{s=1}^n \sum_{t=1}^n \mathrm{Cov}(Z_s, Z_t) = \sigma^2 \sum_{s,t} \rho^{|s-t|}$. Among the n^2 pairs (s,t), exactly n have $|s-t| = 0$ and exactly $2(n-h)$ have $|s-t| = h$ for $1 \le h \le n-1$. Hence

$$\mathrm{Var}\Big(\sum_t Z_t\Big) = \sigma^2\Big[n + 2\sum_{h=1}^{n-1}(n-h)\rho^h\Big],$$

and dividing by n^2 gives the claim.

(b) From (a),

$$n\mathrm{Var}(\bar{Z}_n) = \sigma^2\Big[1 + 2\sum_{h=1}^{n-1}\rho^h - \frac{2}{n}\sum_{h=1}^{n-1} h\rho^h\Big].$$

The first sum tends to $\rho/(1-\rho)$. The series $\sum_h h|\rho|^h$ converges (ratio test), so the last term is at most $\frac{2}{n}\sum_{h\ge 1} h|\rho|^h \to 0$. The limit is $\sigma^2\{1 + 2\rho/(1-\rho)\} = \sigma^2(1+\rho)/(1-\rho)$.

(c) For large n, $\mathrm{Var}(\bar{Z}_n) \approx \sigma^2/n_{\text{eff}}$: the dependent sample is as informative about μ as n_{eff} independent observations. For $\rho = 0.5$, $n_{\text{eff}} = n/3$; for $\rho < 0$ the mean is in fact more precise than under independence. ■

Exercise 1.2 (The nearest neighbor is far away)

Let $X_1, \dots, X_n$ be i.i.d. uniform on the unit ball $B_p = \{x \in \mathbb{R}^p : \|x\| \le 1\}$ and $D = \min_i \|X_i\|$.

(a) Show that the median of D is $d(p,n) = \big(1 - 2^{-1/n}\big)^{1/p}$.

(b) Evaluate $d(10, 500)$ and show that $d(p,n) \to 1$ as $p \to \infty$ for fixed n.

Solution [Supplement].

(a)

Step 1. For $0 \le r \le 1$ the ball rB_p has volume $r^p \,\mathrm{vol}(B_p)$, so $\Pr(\|X_1\| \le r) = r^p$.

Step 2. By independence, $\Pr(D > r) = \prod_i \Pr(\|X_i\| > r) = (1 - r^p)^n$, a continuous, strictly decreasing function of $r \in [0,1]$ from 1 to 0. The median m is therefore the unique solution of $(1 - m^p)^n = 1/2$, i.e. $m^p = 1 - 2^{-1/n}$.

(b) Numerically $d(10, 500) \approx 0.518$: the closest of 500 points is more than halfway to the boundary. For fixed n, $c = 1 - 2^{-1/n} \in (0,1)$ and $c^{1/p} = e^{(\log c)/p} \to e^0 = 1$. In high dimension essentially all points lie near the boundary, far from any target point, so a "nearest" neighbor is not near; this is the curse of dimensionality of the classbox above in another form. ■

Exercise 1.3 (The median cannot be carried away)

Let $n = 2m + 1$ and $x_1, \dots, x_n \in \mathbb{R}$. Replace any $k \le m$ of the observations by arbitrary values and let U be the set of the $n - k$ untouched values, $a = \min U$, $b = \max U$.

(a) Show that the median of the new sample lies in $[a, b]$.

(b) Show that with $m+1$ replaced values the median can be made arbitrarily large, whereas one replaced value suffices for the mean. Conclude that the breakdown points are $(n+1)/(2n) \to 1/2$ for the median and $1/n$ for the mean.

Solution [Supplement].

(a) Let $y_{(1)} \le \cdots \le y_{(n)}$ be the new sample; its median is $y_{(m+1)}$. Every untouched value is $\ge a$, so at most $k \le m$ values are $< a$. If $y_{(m+1)} < a$, then the $m+1$ values $y_{(1)}, \dots, y_{(m+1)}$ would all be $< a$, a contradiction. Likewise at most m values are $> b$; if $y_{(m+1)} > b$, the $n - m = m + 1$ values $y_{(m+1)}, \dots, y_{(n)}$ would all be $> b$, again a contradiction.

(b) Set $m+1$ values equal to $T > \max_i x_i$. The m untouched values are smaller than T, so $y_{(m+1)} = T$, which is arbitrarily large. For the mean, replacing x_1 by $x_1 + t$ moves $\bar{x}$ by t/n, unbounded in t. By (a) no fraction $k/n \le m/n$ breaks the median, so its breakdown point is $(m+1)/n = (n+1)/(2n)$. ■

Exercise 1.4 (Least squares when $p > n$)

Let $X \in \mathbb{R}^{n \times p}$ with $p > n$ and $y \in \mathbb{R}^n$.
(a) Show that $X^\top X$ is singular and that the normal equations $X^\top X\beta = X^\top y$ have infinitely many solutions.
(b) Show that all solutions give the same fitted vector $X\hat{\beta}$; if $\mathrm{rank} X = n$, this vector is y itself (zero training error).
(c) If $\mathrm{rank} X = n$, show that $\beta^\dagger = X^\top (XX^\top)^{-1} y$ is the unique solution of minimum Euclidean norm.

Solution [Supplement].

(a) $\mathrm{rank}(X^\top X) = \mathrm{rank} X \le n < p$, so the $p \times p$ matrix $X^\top X$ is singular and $\mathcal{N}(X) = \{v : Xv = 0\}$ has dimension $\ge p - n \ge 1$. The system is consistent: $\mathrm{col}(X^\top X) \subseteq \mathrm{col}(X^\top)$ and both have the same dimension, so $X^\top y \in \mathrm{col}(X^\top X)$. If $\hat{\beta}$ solves it, so does $\hat{\beta} + v$ for every $v \in \mathcal{N}(X)$, an infinite set.

(b) If β_1, β_2 are solutions, $X^\top X(\beta_1 - \beta_2) = 0$, hence $\|X(\beta_1 - \beta_2)\|^2 = (\beta_1 - \beta_2)^\top X^\top X(\beta_1 - \beta_2) = 0$ and $X\beta_1 = X\beta_2$. If $\mathrm{rank} X = n$, then $\mathrm{col}(X) = \mathbb{R}^n$, so some β has $X\beta = y$; it solves the normal equations, and by the first part every solution has fitted vector y.

(c) $XX^\top$ is $n \times n$ of rank n, hence invertible, and $X\beta^\dagger = y$. The solution set of $X\beta = y$ is $\beta^\dagger + \mathcal{N}(X)$. Since $\beta^\dagger = X^\top w$ lies in $\mathrm{col}(X^\top) = \mathcal{N}(X)^\perp$, for $v \in \mathcal{N}(X)$ we get $\|\beta^\dagger + v\|^2 = \|\beta^\dagger\|^2 + \|v\|^2$, minimized only at $v = 0$.

So with $p > n$ the data cannot pick a coefficient vector, and any of them interpolates the training data: additional structure (sparsity, a penalty) is needed. ■